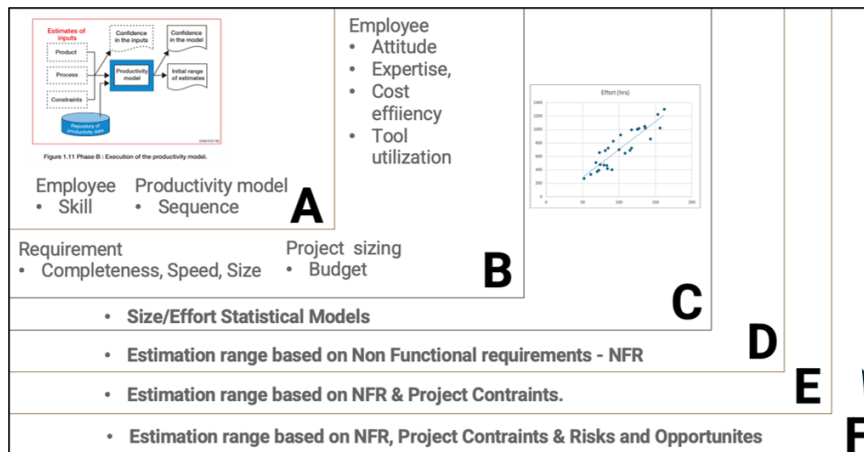


Software Projects Estimation (Educational Games)

Module F: Selecting a Budget and a Contingency

(Version: July 15, 2026)

Audience: Students



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Module F:

Selecting a Budget and a Contingency

Using Risks and Opportunities to Select a Budget and a Contingency

1. Introduction

This module is designed to teach students how to select a budget and a related contingency reserve by taking into account the risks and opportunities identified at estimation time.

For this purpose, this educational game provides three main scenarios with nine sub-scenarios of risks and opportunities.

2. Learning Objectives

Teach students how to **integrate a set of identified risks and opportunities** into the selection of a **project budget and an associated contingency reserve**, taking into account the risk attitude of the project manager.

What the player sees:

- A regression model graph with equation line and $\pm$ MMRE bounds for a 500 CFP project
- A reference table of 7 risks with their effort impact percentages (Low / Medium / High)
- A reference table of 3 opportunities with their effort impact percentages (Low / Medium / High)
- A probability scale ranging from Remotely to Near certainty
- Three scenarios, each specifying an estimation range, a manager profile, and assigned impact and probability levels for each risk and opportunity

How the game works:

1. Select a manager profile (High risk-taking / On-target / Balanced).
2. Select one of the three available scenarios for that profile.
3. Answer 5 sequential budget questions using the questionnaire data.
4. Return to profile and scenario selection; repeat for additional combinations.
5. Trigger **EndGame** at any time after completing at least three scenarios.

Once a scenario is completed under a given profile, it becomes unavailable for that profile. The same scenario remains accessible under the other profiles.

9 Scenario–profile combinations : 3 scenarios × 3 manager profiles

- **Input:** An estimation range, 7 risks, 3 opportunities, and a manager risk profile.
- **Action:** Calculate combined impact and determine base budget, project and corporate contingency reserves.
- **Outcome:** Immediate scoring across 5 sequential budget questions per scenario.

3. Prerequisites

A student (e.g., a game player) should have a basic understanding of software engineering and management perspective and of the previous educational games in this set on Software Projects Estimation.

4. Time Commitment

The average time to play a single game is 10-15 minutes, but, of course, it is likely to take several iterations playing the game for the student to learn the concepts and be able to answer the questions. Participants should be given at least one week of out-of-class time to explore the game and answer the questions (see Section 8).

5. Instructions

<https://chairegestionproduitsprojetslogiciels.github.io/simse-software-estimation-capability/games/game-f-risks-budget.html>

The game also includes a **questionnaire** that provide detailed instructions for using the simulation. If Java is not installed on your computer, you may be prompted to install it when launching the JAR version. If Java is not already installed on your computer, follow the Java installation instructions provided in the **Resources** section before launching the JAR version.

6. Questions

Choose two to three of the following questions to use with this module.

1. In terms of impact of project effort, what is the difference between a risk and an opportunity?
2. In previous modules, a range of estimated effort has been identified based on functional requirements, non-functional requirements as well on project constraints. How can you consider risks and opportunity for making a decision for a specific budget?
3. Explain the risks profiles of project managers and how this influences the allocation of a budget to a project.
4. When a project manager has selected the top line of the estimated budget, how much corporate contingency will he be offered? Will it be zero?

7. Feedback?

Any comments, suggestions, feedback, or experience regarding this module?

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References

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